[의약정보학]

PubMed 사용하기

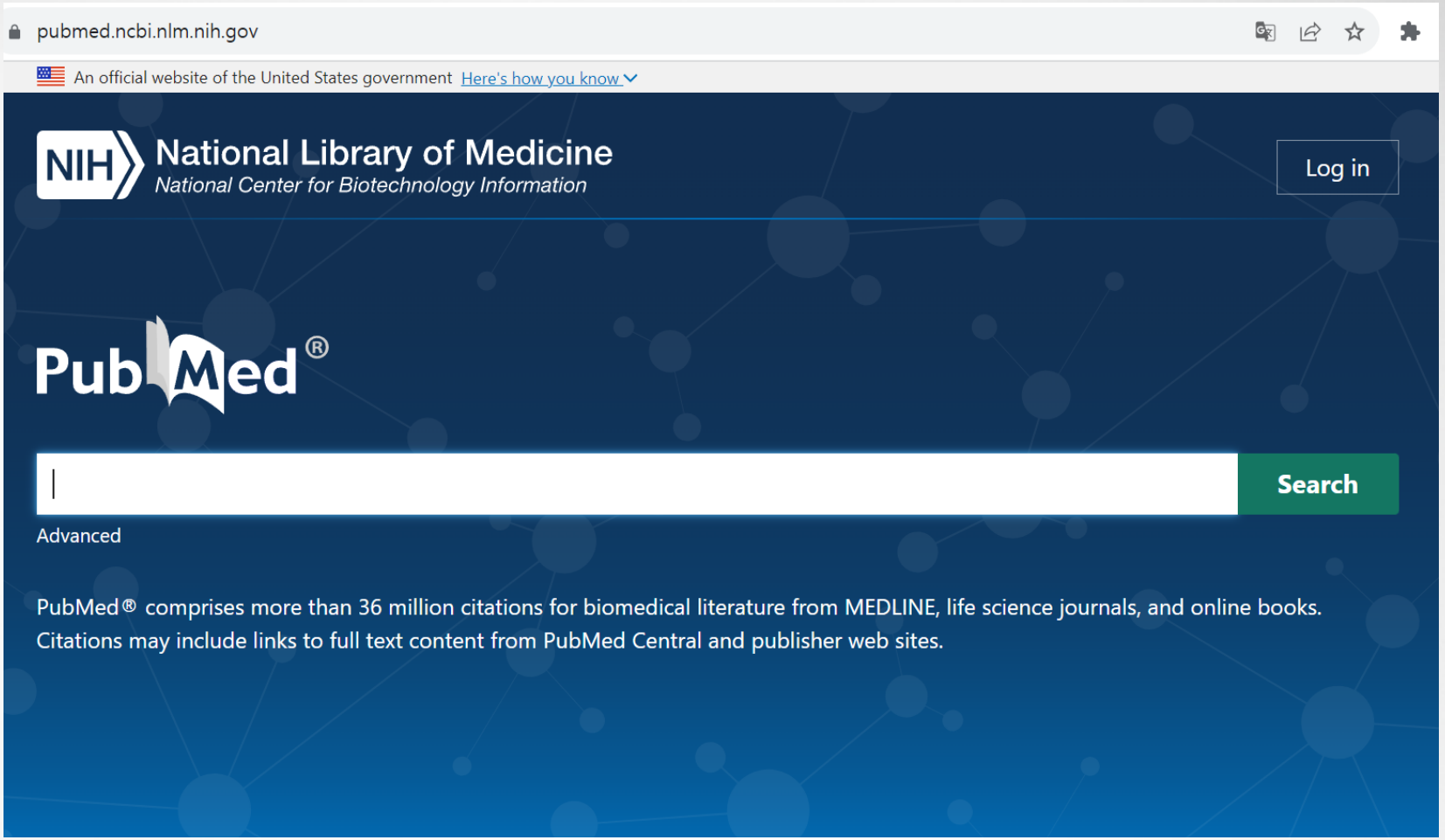
Kyungim Kim, Ph.D., MHPE
College of Pharmacy, Korea University

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1. PubMed 개요

- **의학 및 생명과학 분야 문헌 검색 지원 무료 데이터베이스**



1. PubMed 개요

- **의학 및 생명과학 분야 문헌 검색 지원 무료 데이터베이스**

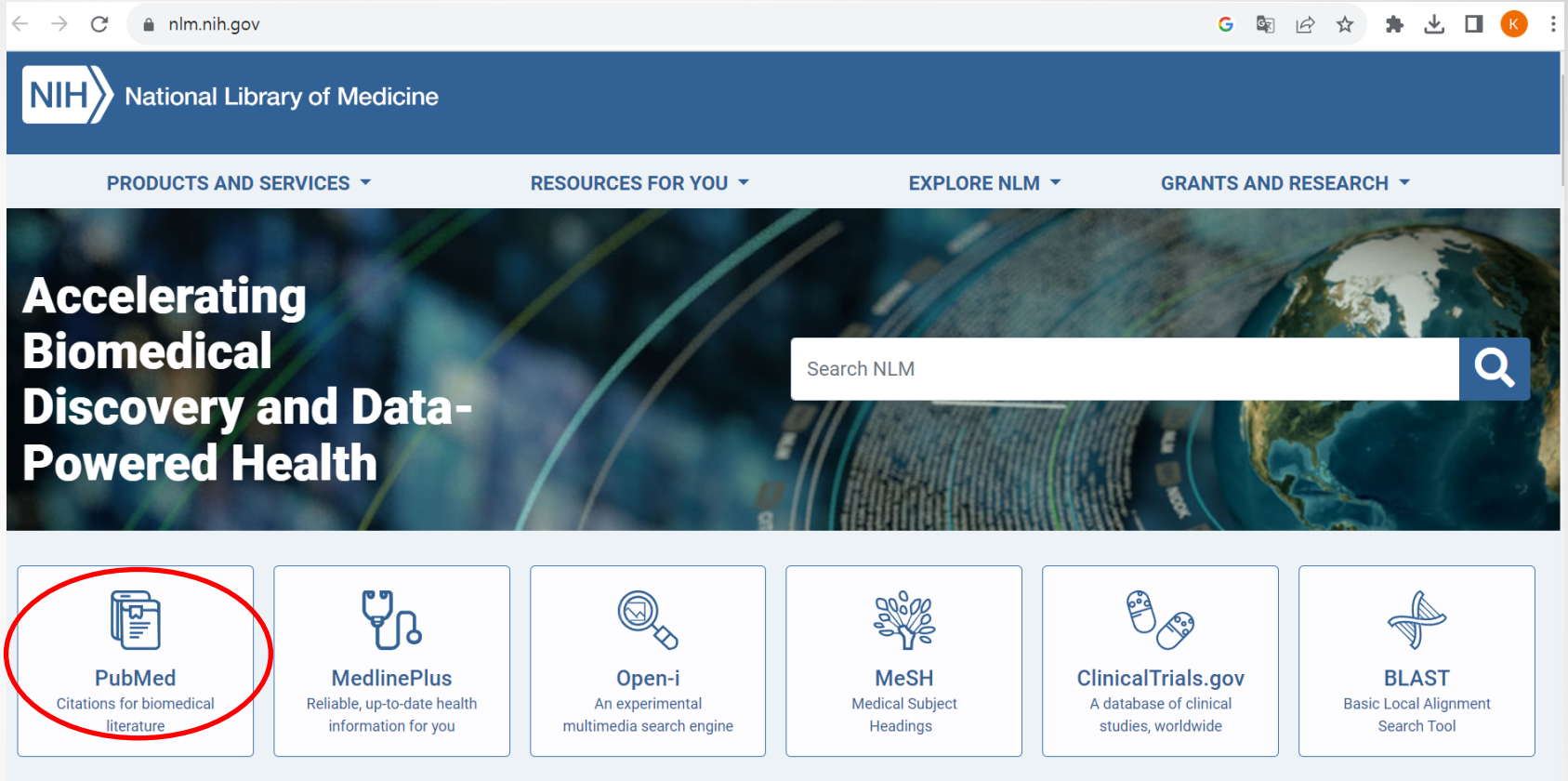
  a. 미국 국립의학도서관(National Library of Medicine, NLM) 산하 국립생물공학정보센터(the National Center for Biotechnology Information, NCBI)가 운영

  b. 의학, 약학, 간호학, 치의학, 수의학 등 의약학, 생명과학, 행동과학, 화학, 의공학 등 관련 학술지의 서지정보(색인 초록) DB



1. PubMed 개요

- 의학 및 생명과학 분야 문헌 검색 지원 무료 데이터베이스



1. PubMed 개요

- **의학 및 생명과학 분야 문헌 검색 지원 무료 데이터베이스**

c. 생물의학 분야 4천만건 논문의 서지, 초록, 인용 정보 수록

- 전세계 40개 언어(영어 논문 86%), 1946 년 이후
- 2011년부터 매년 1백만 건 이상 증가, 매일 업데이트

d. 데이터베이스 구성

- MEDLINE(NLM의 의학 및 생명과학 분야 서지 데이터베이스) 등재학술지 약 5,300종(전체 논문의 80% 이상)
- PubMed Central (PMC) 원문제공 학술지 약 3,300종 (이중 MEDLINE 등재는 약 900종)
- 그 외 non-MEDLINE 학술지, Bookshelf (e-book chapter), Preprint (NIH 펀딩 연구) 등

1. PubMed 개요

의학 및 생명과학 분야 문헌 검색 지원 무료 데이터베이스

DB명	설 명	학술지 종수	수록 및 검색연도	서지 정보	초록 정보	참고 문헌	피인용 횟수	원문 링크
PubMed	_ Medline (OLD Medline 포함) Old Medline(1946-65)+Medline(1966-현재) _ 의학, 보건학 등 생의학 관련 5,264 학술지의 논문 서지/초록 수록, 검색 [미국 중심 의학 대표 DB, 40개 언어] _ MeSH (Medical Subject Headings) 색인 _ Medline DB는 제공사 플랫폼으로 다양하게 검색 가능 [OVID, EBSCO, Proquest..]	5,264*	1946-	v	v			v
	_ PMC (PubMed Central) _ NIH 생의학관련 무료 디지털 아카이브 저장소로 OA 협력 출판사/일부 기사 및 저자 논문(NIH 기금)의 원문 제공	3,625**	2000-	v	v	v	v	원문 수록
	_ NCBI Bookshelf _ 'NCBI Bookshelf'(NLM 도서정보) 에서 이용할 수 있는 도서 인용 및 원문 정보(일부 Book chapter 포함) 또는 기술보고서, 참고도서, 가이드라인 등 수록	8,429**	2010-	v		v		원문 수록

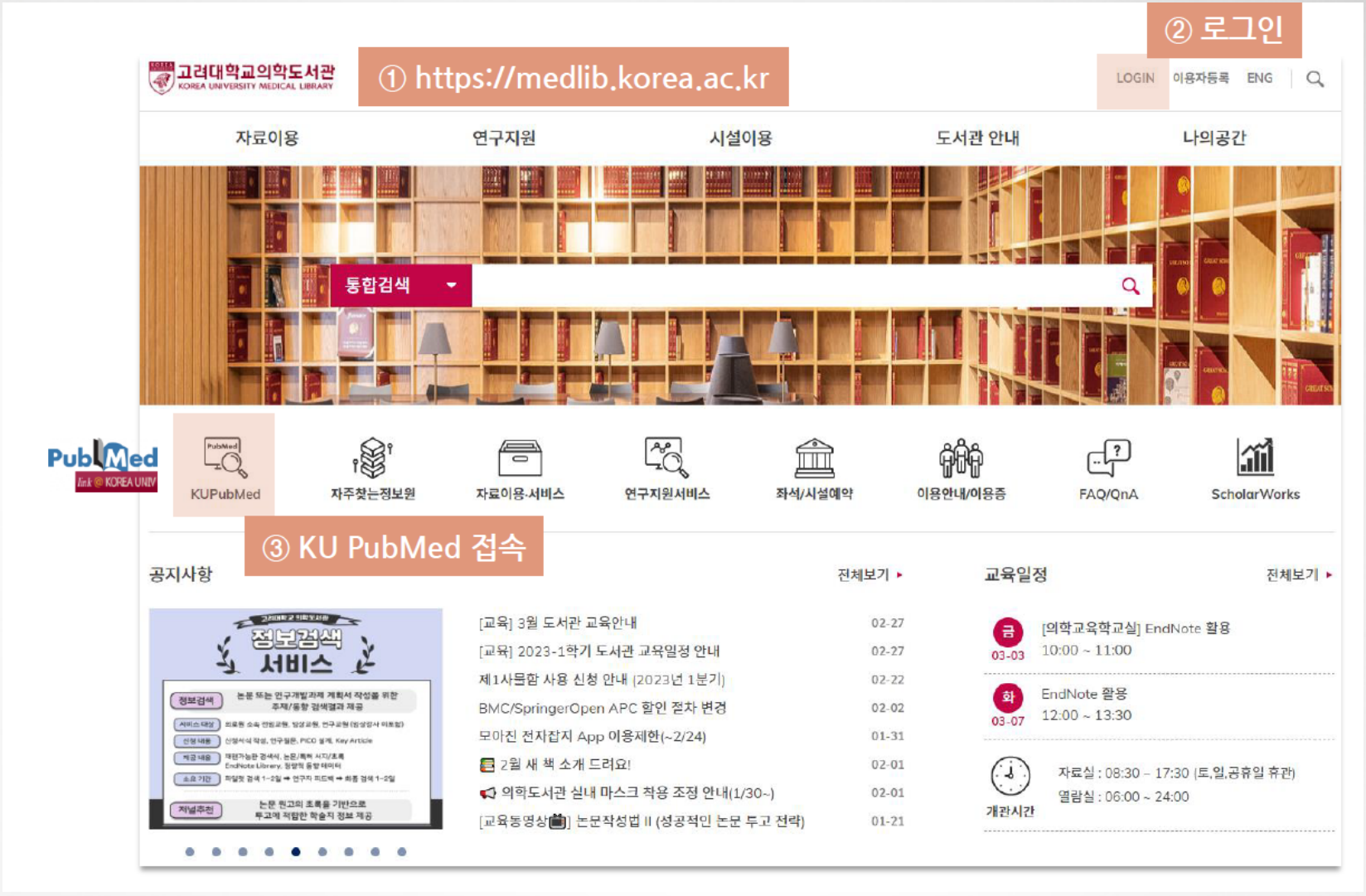
*미국 국립의학도서관의 LSTRC (Literature Selection Technical Review Committee)에서 선정
- 주제범위, 내용 편집 간행물의 질, 대상독자 등 고려 [https://www.ncbi.nlm.nih.gov/nlmcatalog?term=currentlyindexed]
**https://www.ncbi.nlm.nih.gov/pmc/journals/, https://www.ncbi.nlm.nih.gov/books/browse/

1. PubMed 개요

- PubMed 접근
  a. https://pubmed.gov
  b. https://pubmed.ncbi.nlm.nih.gov
  c. 고려대학교 의학도서관 KU PubMed를 통한 접근

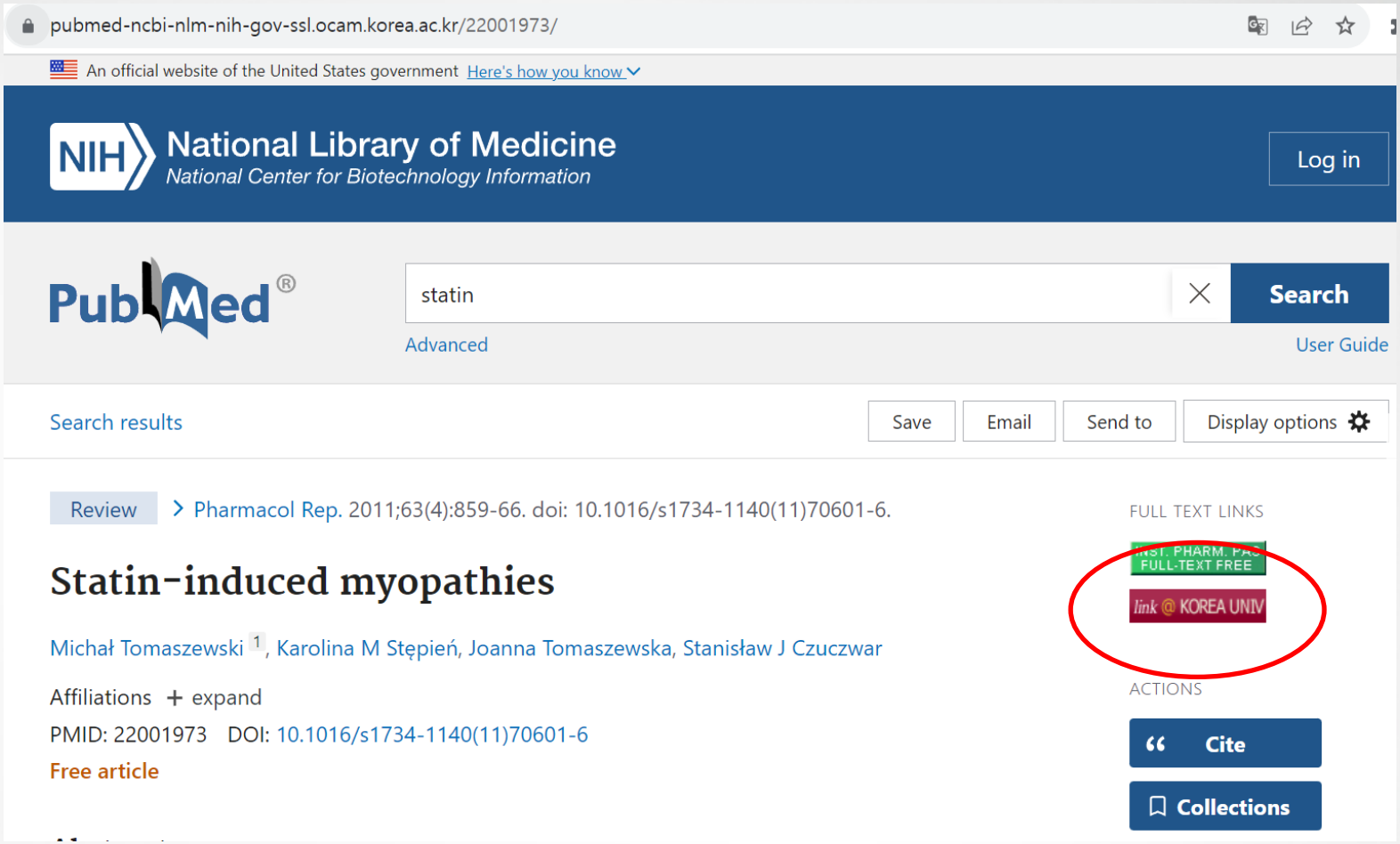
1. PubMed 개요

- PubMed 접근



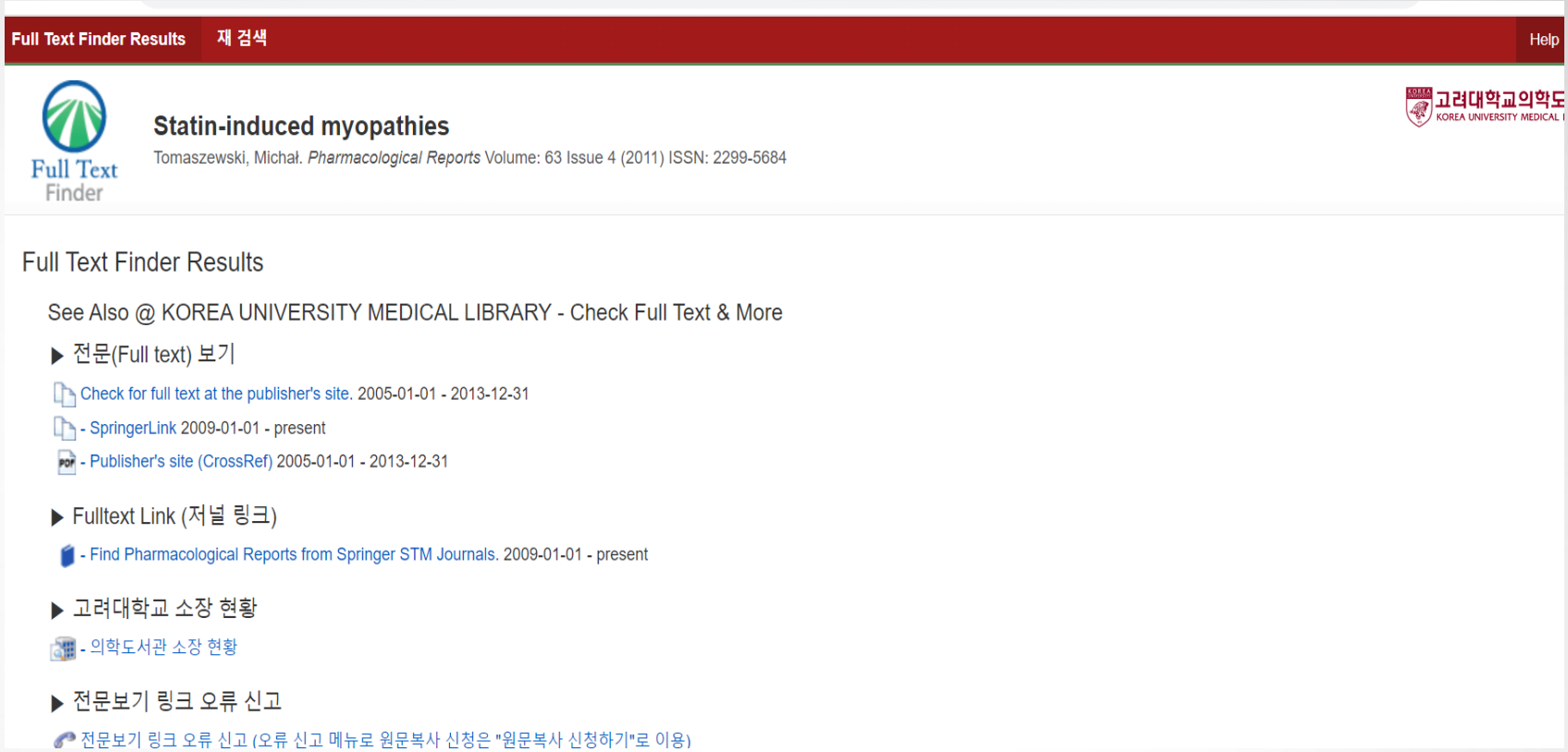
1. PubMed 개요

- PubMed 접근



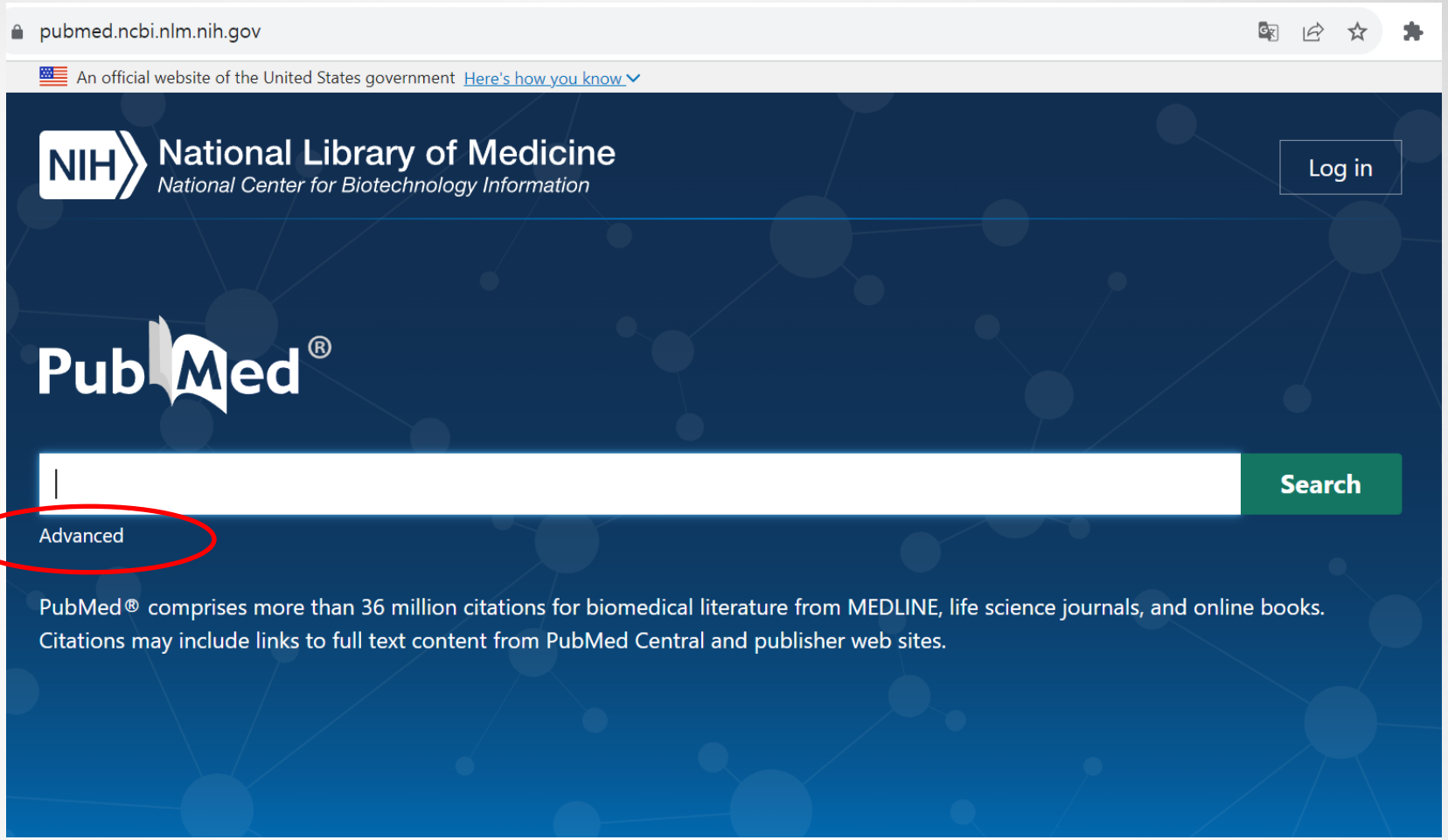
1. PubMed 개요

- PubMed 접근



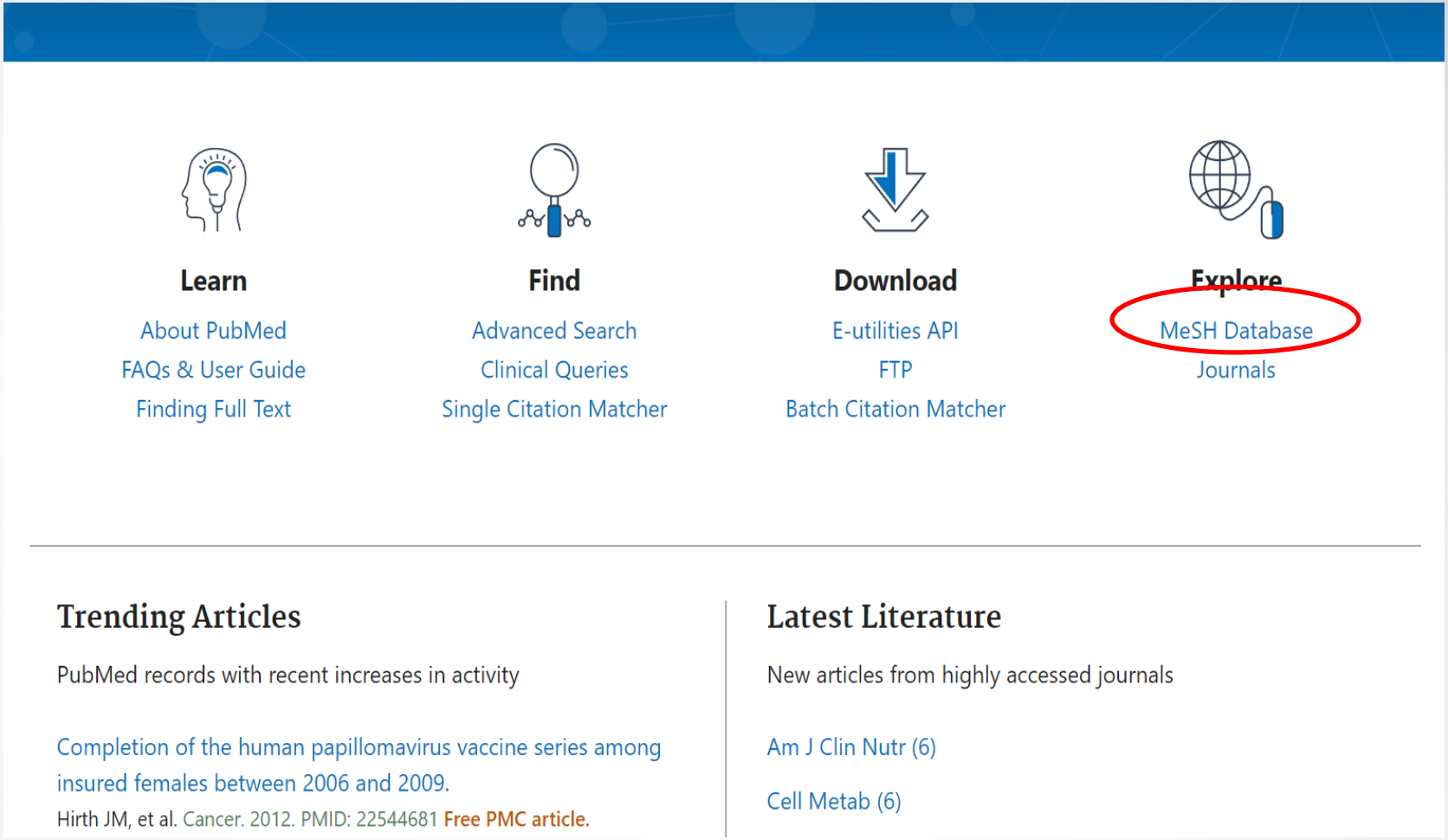
1. PubMed 개요

PubMed 살펴보기



1. PubMed 개요

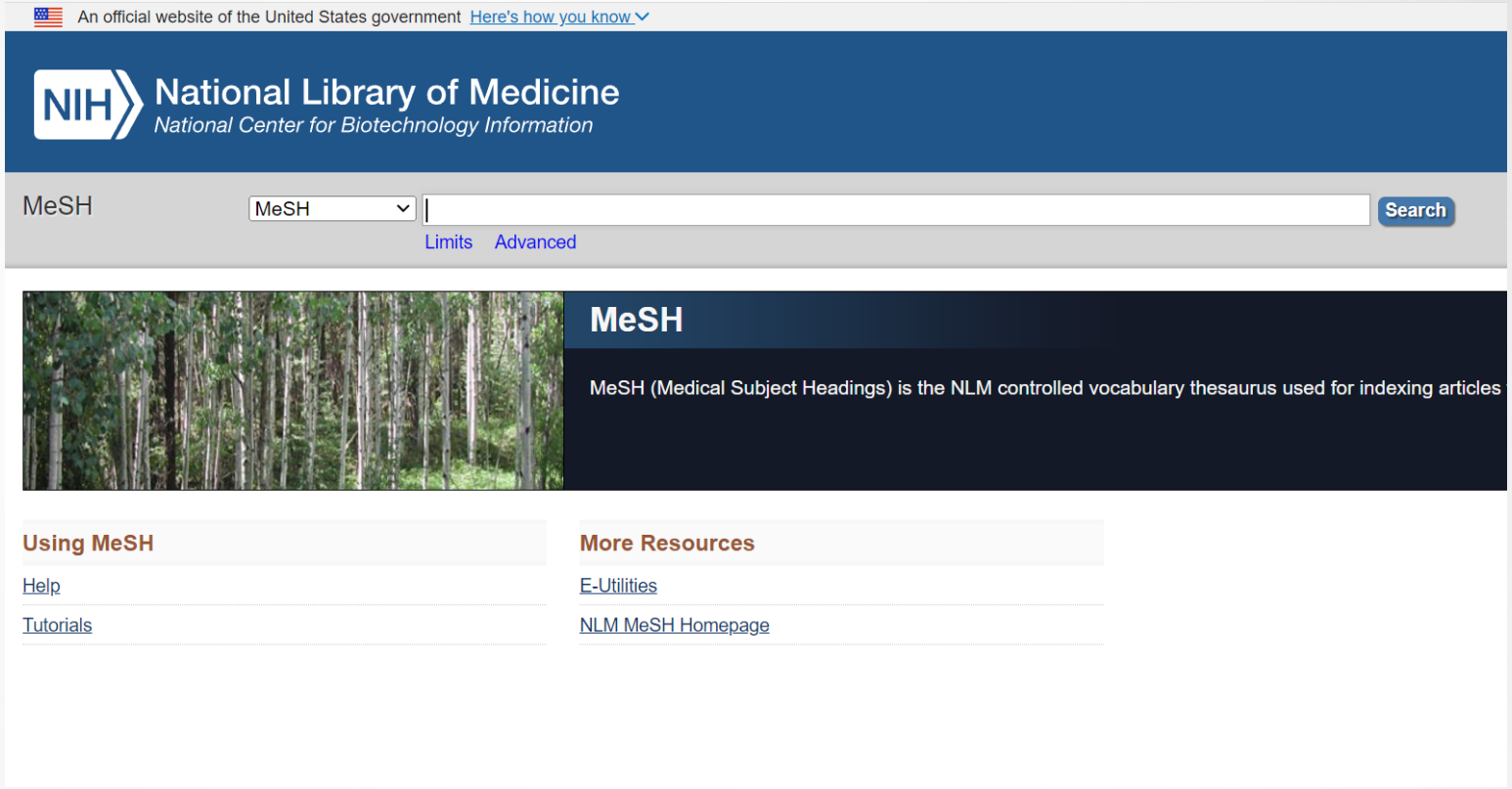
- **PubMed 살펴보기**



1. PubMed 개요

- **The Medical Subject Headings (MeSH)**

  a. PubMed의 서지정보 색인을 생성하는 데 사용되는 NLM의 통제된 어휘 동의어집



1. PubMed 개요

- **The Medical Subject Headings (MeSH)**

  b. 상위/하위 개념을 분류한 16개 트리 구조

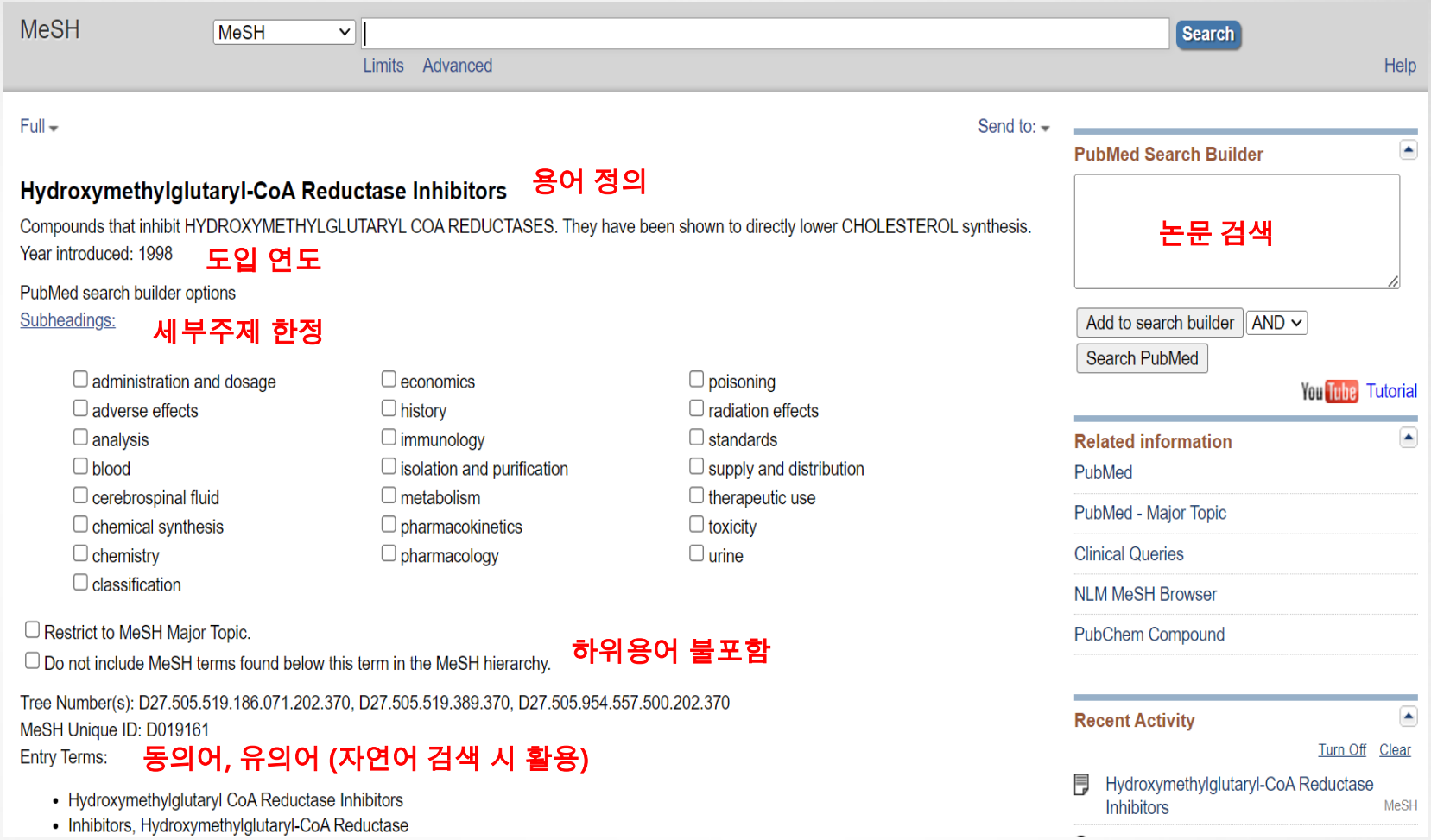


  c. 연1회 개정, 신규용어 포함

  - 예) COVID-19, 2021(2020) 도입; Semaglutide(2026 도입)

1. PubMed 개요

- The Medical Subject Headings (MeSH)



1. PubMed 개요

- The Medical Subject Headings (MeSH)

Previous Indexing: 2007년 이전 색인용어

- Enzyme Inhibitors (1976-1996)
- Hydroxymethylglutaryl CoA Reductases/antagonists and inhibitors (1976-1996)

See Also:

- Hydroxymethylglutaryl CoA Reductases 관련 용어

All MeSH Categories
Chemicals and Drugs Category
Chemical Actions and Uses
Pharmacologic Actions
Molecular Mechanisms of Pharmacological Action
Antimetabolites
Hypolipidemic Agents
Anticholesteremic Agents
Hydroxymethylglutaryl-CoA Reductase Inhibitors

MeSH 트리구조
(용어 상하위 구조/계층 참고)

All MeSH Categories
Chemicals and Drugs Category
Chemical Actions and Uses
Pharmacologic Actions
Molecular Mechanisms of Pharmacological Action
Enzyme Inhibitors
Hydroxymethylglutaryl-CoA Reductase Inhibitors

All MeSH Categories
Chemicals and Drugs Category
Chemical Actions and Uses
Pharmacologic Actions
Therapeutic Uses
Lipid Regulating Agents
Hypolipidemic Agents

2. PubMed 검색

- **검색 전략 수립**

  a. 원하는 정보를 찾기 위해 명확한 질문을 설정

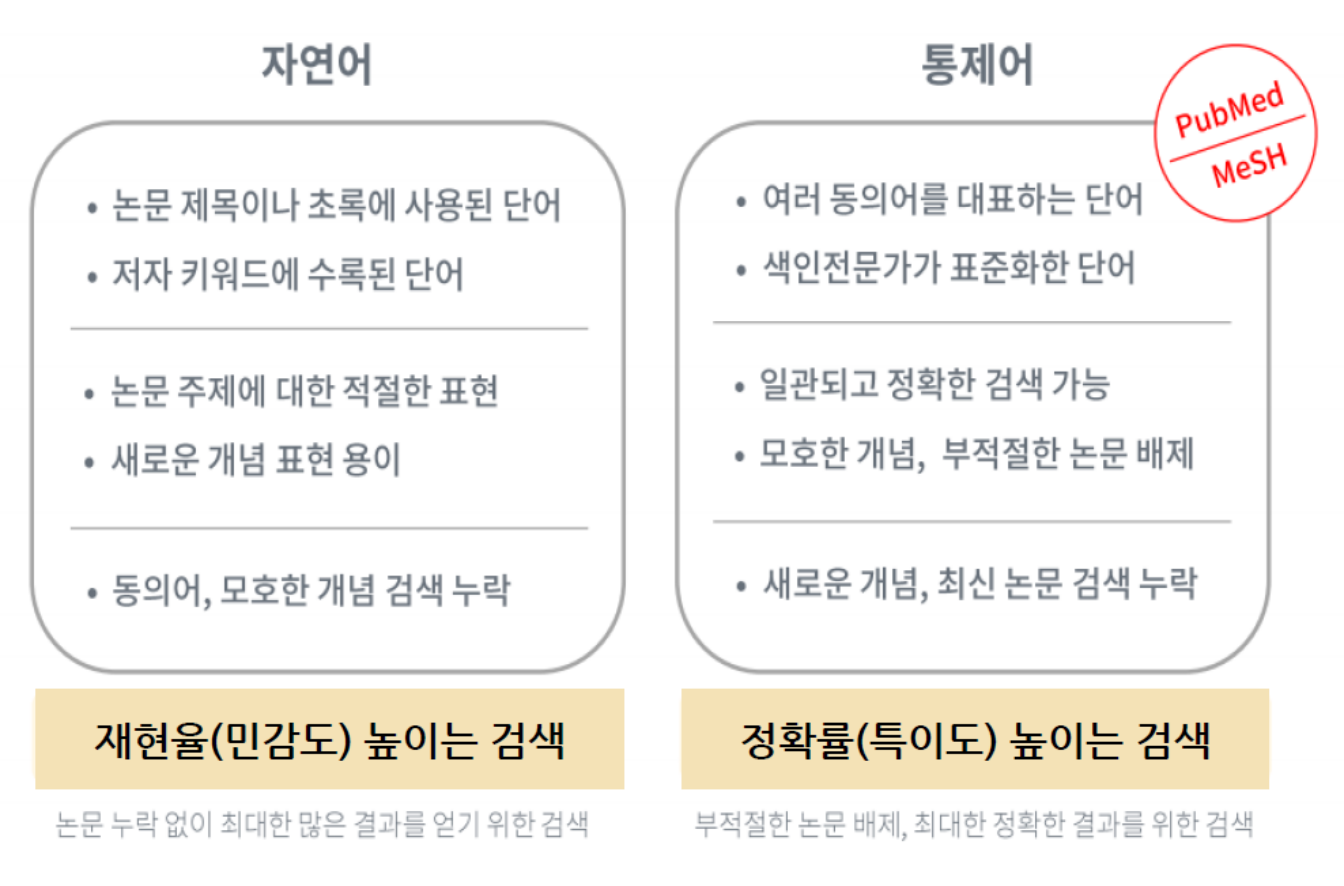
  b. Tips

  - Identify the key concepts.
  - Determine alternative terms for these concepts, if needed.
  - Refine your search to dates, study groups, etc., as appropriate.
  - Practice helps. Strategies and styles will differ according to personal choice and professional discipline.

2. PubMed 검색

- **검색 전략 수립**

  c. 자연어 vs. MeSH

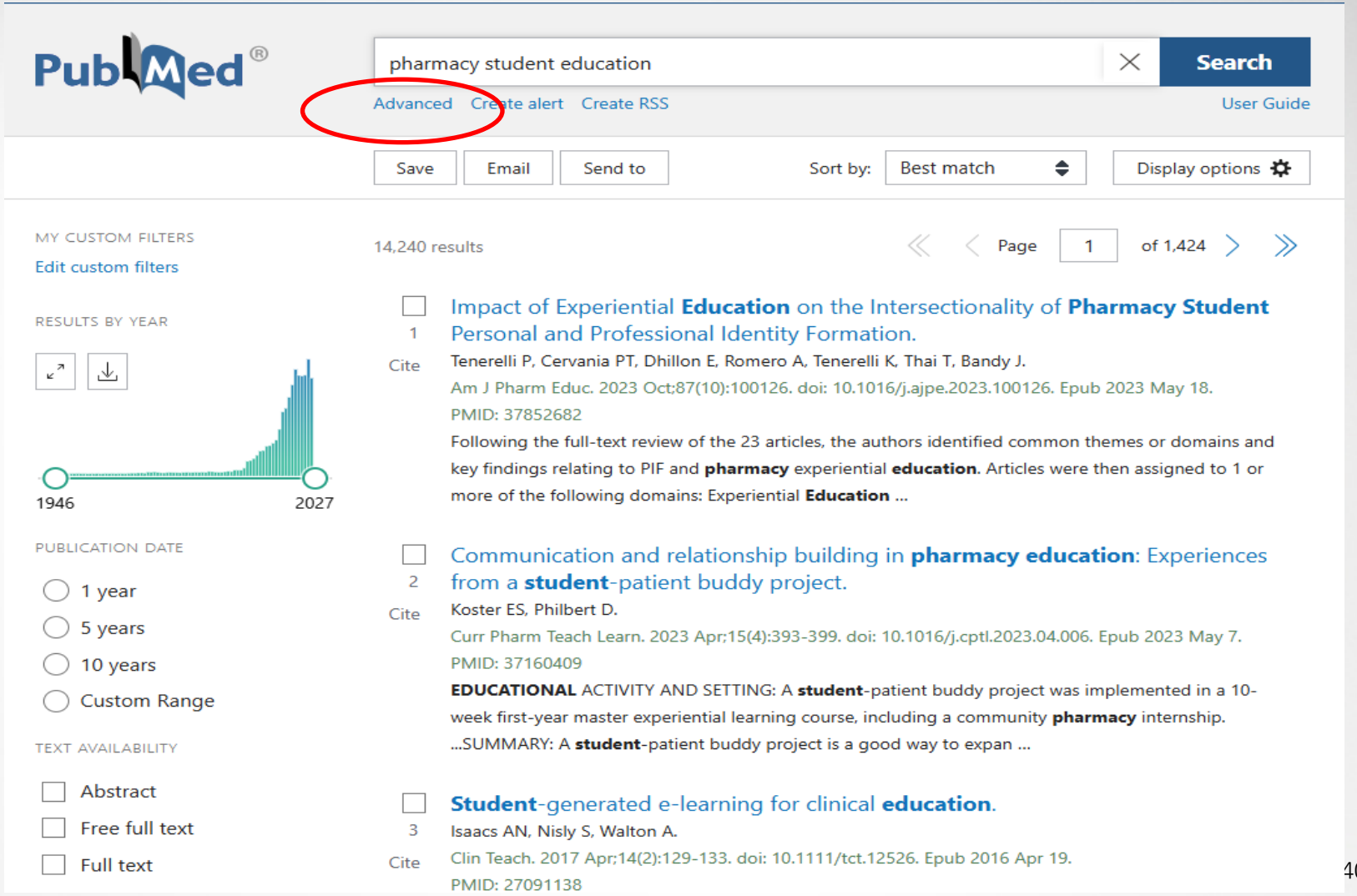


2. PubMed 검색

- **Automatic term mapping**
  a. 검색어를 입력하면 MeSH 〉 Journal 〉 Author 순으로 매핑하고 변환하여 검색
  b. 변환 상세내용 확인
     - Advanced → Details

2. PubMed 검색

- Automatic term mapping



2. PubMed 검색

- **Automatic term mapping**

  c. MeSH OR All Fields (통제어 OR 자연어)

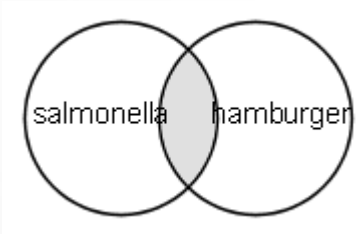
Search	Actions	Details	Query	Results	Time
#2	•••	⌄	Search: **pharmacy student education** ("students, pharmacy"[MeSH Terms] OR ("students"[All Fields] AND "pharmacy"[All Fields]) OR "pharmacy students"[All Fields] OR ("pharmacy"[All Fields] AND "student"[All Fields]) OR "pharmacy student"[All Fields]) AND ("educability"[All Fields] OR "educable"[All Fields] OR "educates"[All Fields] OR "education"[MeSH Subheading] OR "education"[All Fields] OR "educational status"[MeSH Terms] OR ("educational"[All Fields] AND "status"[All Fields]) OR "educational status"[All Fields] OR "education"[MeSH Terms] OR "education s"[All Fields] OR "educational"[All Fields] OR "educative"[All Fields] OR "educator"[All Fields] OR "educator s"[All Fields] OR "educators"[All Fields] OR "teaching"[MeSH Terms] OR "teaching"[All Fields] OR "educate"[All Fields] OR "educated"[All Fields] OR "educating"[All Fields] OR "educations"[All Fields]) **Translations** **pharmacy student:** "students, pharmacy"[MeSH Terms] OR ("students"[All Fields] AND "pharmacy"[All Fields]) OR "pharmacy students"[All Fields] OR ("pharmacy"[All Fields] AND "student"[All Fields]) OR "pharmacy student"[All Fields] **education:** "educability"[All Fields] OR "educable"[All Fields] OR "educates"[All Fields] OR "education"[Subheading] OR "education"[All Fields] OR "educational status"[MeSH Terms] OR ("educational"[All Fields] AND "status"[All Fields]) OR "educational status"[All Fields] OR "education"[MeSH Terms] OR "education's"[All Fields] OR "educational"[All Fields] OR "educative"[All Fields] OR "educator"[All Fields] OR "educator's"[All Fields] OR "educators"[All Fields] OR "teaching"[MeSH Terms] OR "teaching"[All Fields] OR "educate"[All Fields] OR "educated"[All Fields] OR "educating"[All Fields] OR "educations"[All Fields]	14,240	08:53:49

2. PubMed 검색

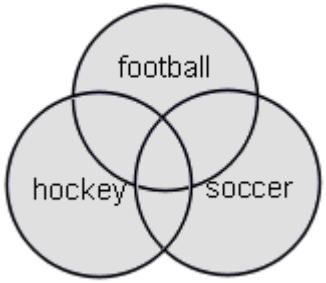
- **검색 기술**

a. Boolean logic

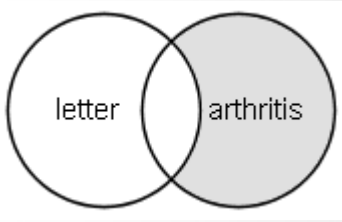
- 검색어 간의 논리적 관계 설정
- AND (space ; default)
- OR
- NOT



salmonella AND hamburger



football OR hockey OR soccer



arthritis NOT letter

2. PubMed 검색

- **검색 기술**

  b. 우선순위(Nesting)

  - ( ) 사용하여 우선 검색
  - 왼쪽에서 오른쪽으로



salmonella AND hamburger OR eggs

salmonella AND (hamburger OR eggs)

2. PubMed 검색

- **검색 기술**

  c. Truncation(절단검색)

  - '*' 사용 (* 앞에 최소 4글자가 필요)
  - 예: pharmac*

Actions	Details	Query	Results	Time
•••	⌄	Search: **pharmacy** Sort by: **Publication Date** "pharmacie"[All Fields] OR "pharmacies"[MeSH Terms] OR "pharmacies"[All Fields] OR "pharmacy"[MeSH Terms] OR "pharmacy"[All Fields] OR "pharmacy s"[All Fields] **Translations** **pharmacy:** "pharmacie"[All Fields] OR "pharmacies"[MeSH Terms] OR "pharmacies"[All Fields] OR "pharmacy"[MeSH Terms] OR "pharmacy"[All Fields] OR "pharmacy's"[All Fields]	688,993	10:29:22
•••	⌄	Search: **pharmac*** Sort by: **Publication Date** "pharmac*"[All Fields]	5,443,047	10:29:03

2. PubMed 검색

- **검색 기술**

d. Field Tags

- 논문이 너무 많거나 부정확한 결과가 나올 때, 검색 범위를 제한

Affiliation [ad]	Full Investigator Name [fir]	Pagination [pg]
All Fields [all]	Grant Number [gr]	Personal Name as Subject [ps]
Article Identifier [aid]	Investigator [ir]	Pharmacological Action [pa]
Author [au]	ISBN [isbn]	Place of Publication [pl]
Author Identifier [auid]	Issue [ip]	PMCID and MID
Book [book]	Journal [ta]	PMID [pmid]
Comment Correction Type	Language [la]	Publication Date [dp]
Completion Date [dcom]	Last Author Name [lastau]	Publication Type [pt]
Conflict of Interest Statement [cois]	Location ID [lid]	Publisher [pubn]
Corporate Author [cn]	MeSH Date [mhda]	Secondary Source ID [si]
Create Date [crdt]	MeSH Major Topic [majr]	Subset [sb]
EC/RN Number [rn]	MeSH Subheadings [sh]	Supplementary Concept [nm]
Editor [ed]	MeSH Terms [mh]	Text Words [tw]
Entry Date [edat]	Modification Date [lr]	Title [ti]
Filter [filter] [sb]	NLM Unique ID [jid]	Title/Abstract [tiab]
First Author Name [1au]	Other Term [ot]	Transliterated Title [tt]
Full Author Name [fau]	Owner	Volume [vi]

2. PubMed 검색

- **검색 기술**

  d. Field Tags

  - 검색어마다 필드 태그를 붙여야 함
  - 예1) caffeine[mh] OR caffein*[tiab]
  - 예2) covid-19[tw] NOT preprint[pt]
  - 예3) muscle strength[majr] AND 2010:2023[dp]
  - 예4) caffeine[ti] OR coffee[ti] (O) / (caffeine OR coffee)[ti] (X)

2. PubMed 검색

- **검색 기술**

  e. Phrase searching (어구검색)

  - 앞뒤에 큰따옴표 (“ ”) 를 사용하여 하나의 어구로 검색

Search	Actions	Details	Query	Results	Time
#19	•••	⌄	Search: **"pharmacy students"** Sort by: **Publication Date** "pharmacy students"[All Fields]	5,941	10:41:39
#18	•••	⌄	Search: **pharmacy students** Sort by: **Publication Date** "students, pharmacy"[MeSH Terms] OR ("students"[All Fields] AND "pharmacy"[All Fields]) OR "pharmacy students"[All Fields] OR ("pharmacy"[All Fields] AND "students"[All Fields]) **Translations** **pharmacy students:** "students, pharmacy"[MeSH Terms] OR ("students"[All Fields] AND "pharmacy"[All Fields]) OR "pharmacy students"[All Fields] OR ("pharmacy"[All Fields] AND "students"[All Fields])	11,969	10:41:11

2. PubMed 검색

- **검색 기술**

f. Proximity searching (인접검색)

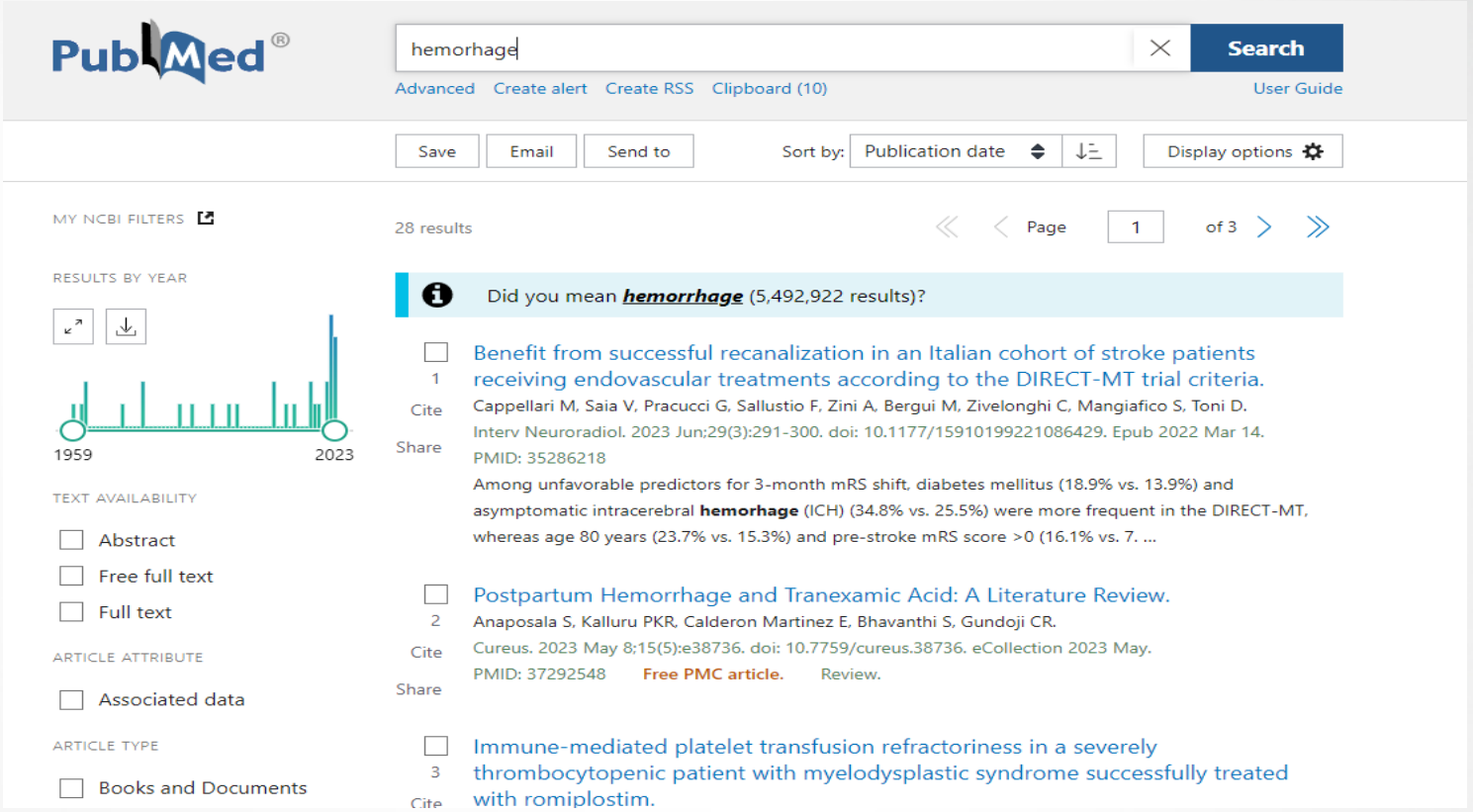
- “검색어” [field:~N]
  : 검색어 = 2 개 이상의 단어를 큰따옴표 (“ “) 안에 기술
  : Field = [title] 또는 [title/abstract] 필드에서 가능
  : N = 검색어 사이에 N 개 이하 단어가 포함된 경우 검색
- 어구검색이 되지 않는 경우, N을 0으로 하여 검색 가능
- 단어 순서는 관계 없음
  : "pharmacy education" [ti:~2] vs. "pharmacy education" [ti:~0]

Search	Actions	Details	Query	Results	Time
#26	•••	›	Search: **"pharmacy education" [ti:~0]** Sort by: **Publication Date**	642	10:50:14
#25	•••	›	Search: **"pharmacy education" [ti:~2]** Sort by: **Publication Date**	896	10:49:52

2. PubMed 검색

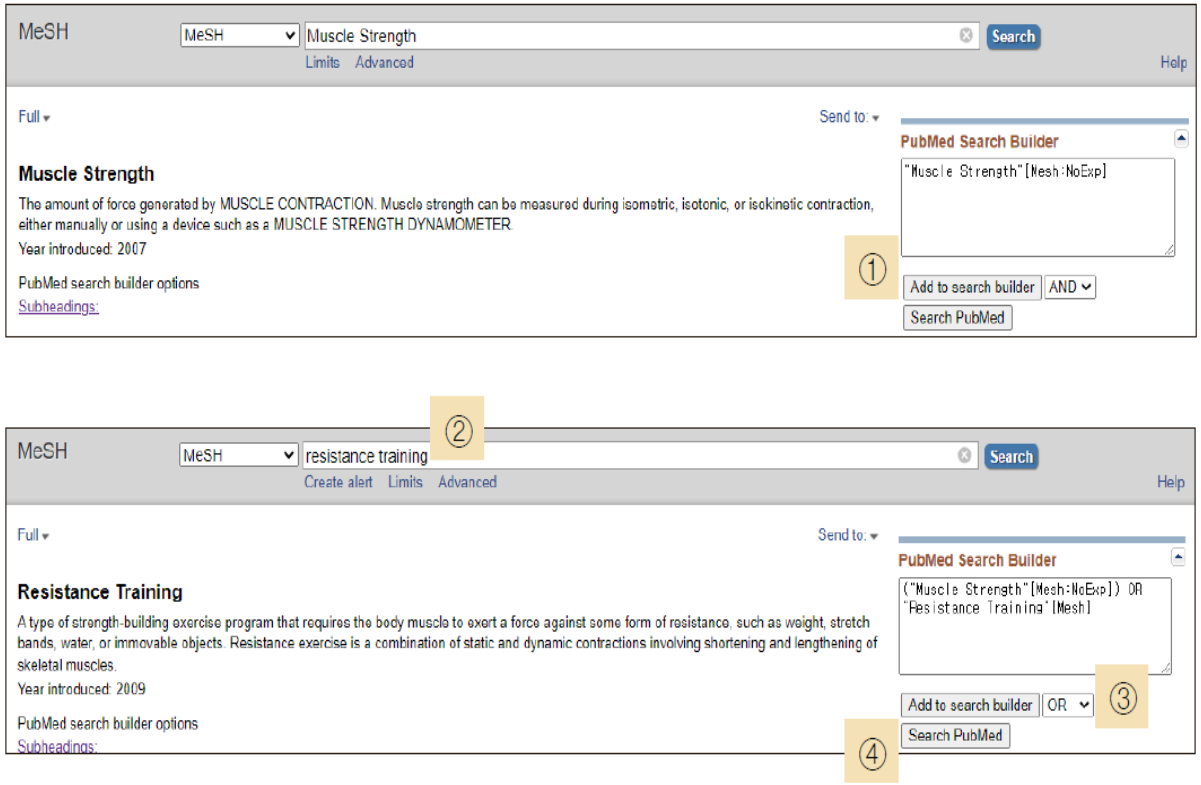
- **검색 기술**

  g. 스펠링 오류 확인 기능



2. PubMed 검색

MeSH DB를 이용한 검색



① 세부주제/중심주제 한정, 하위 용어 불포함 한정 등 원하는 옵션을 체크한 후 Add to search builder 클릭

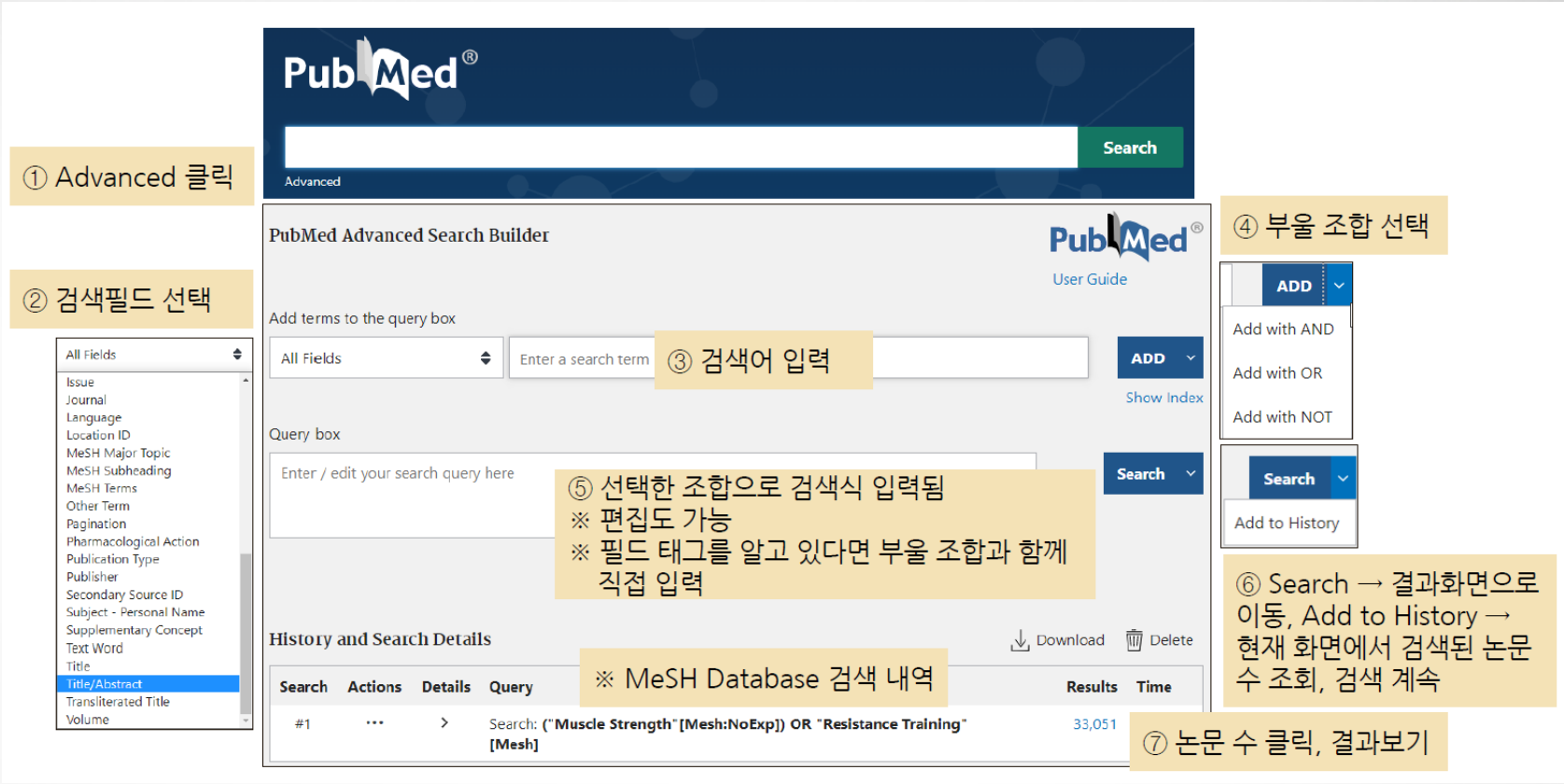
② Search PubMed 클릭 또는 다른 MeSH 용어 검색

③ AND/OR/NOT 원하는 조합을 선택하고 Add to search builder 클릭

④ Search PubMed 클릭

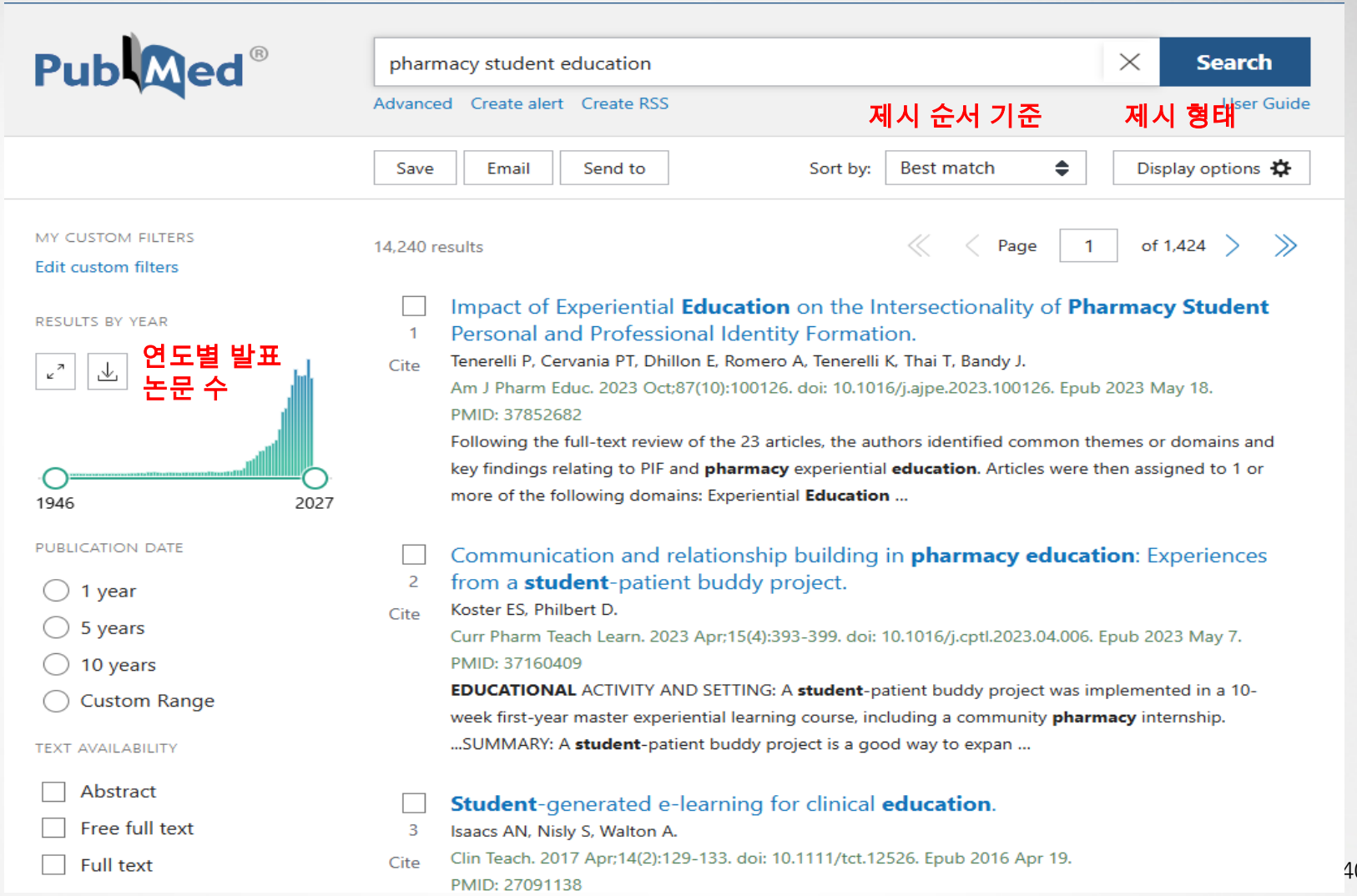
2. PubMed 검색

Advanced 조합 검색



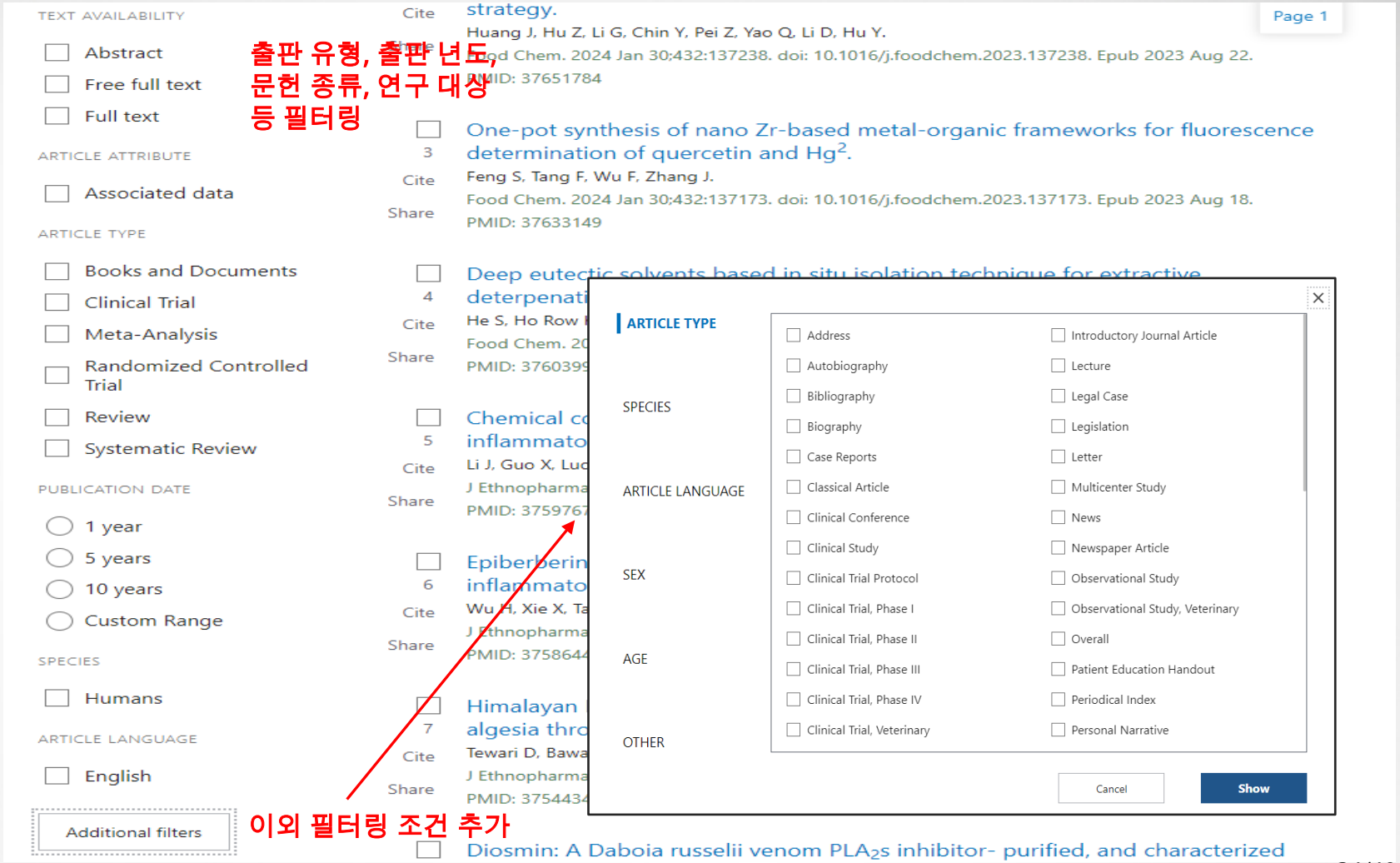
3. PubMed 결과보기

- **검색 결과 살펴보기**



3. PubMed 결과보기

- **검색 결과 살펴보기: Filter 기능 활용**

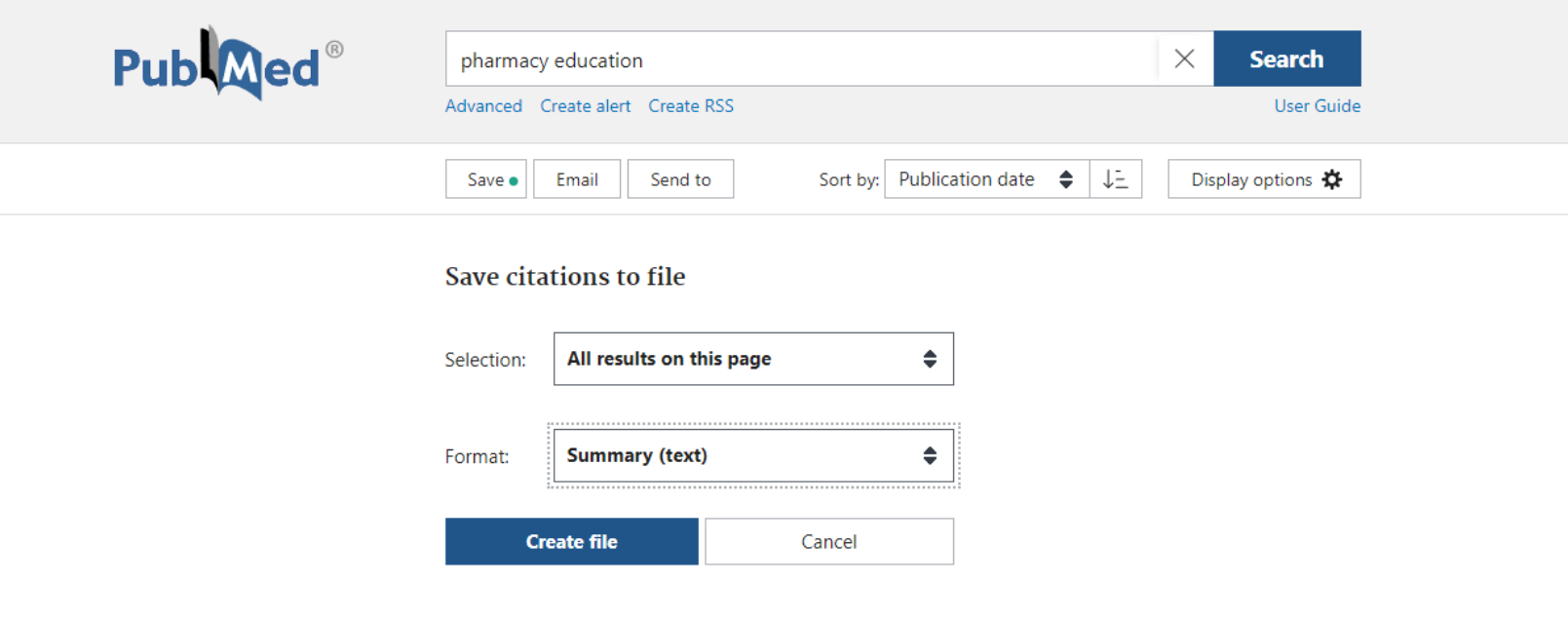


3. PubMed 결과보기

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a. Save 기능

- Text 또는 CSV 파일로 다운로드

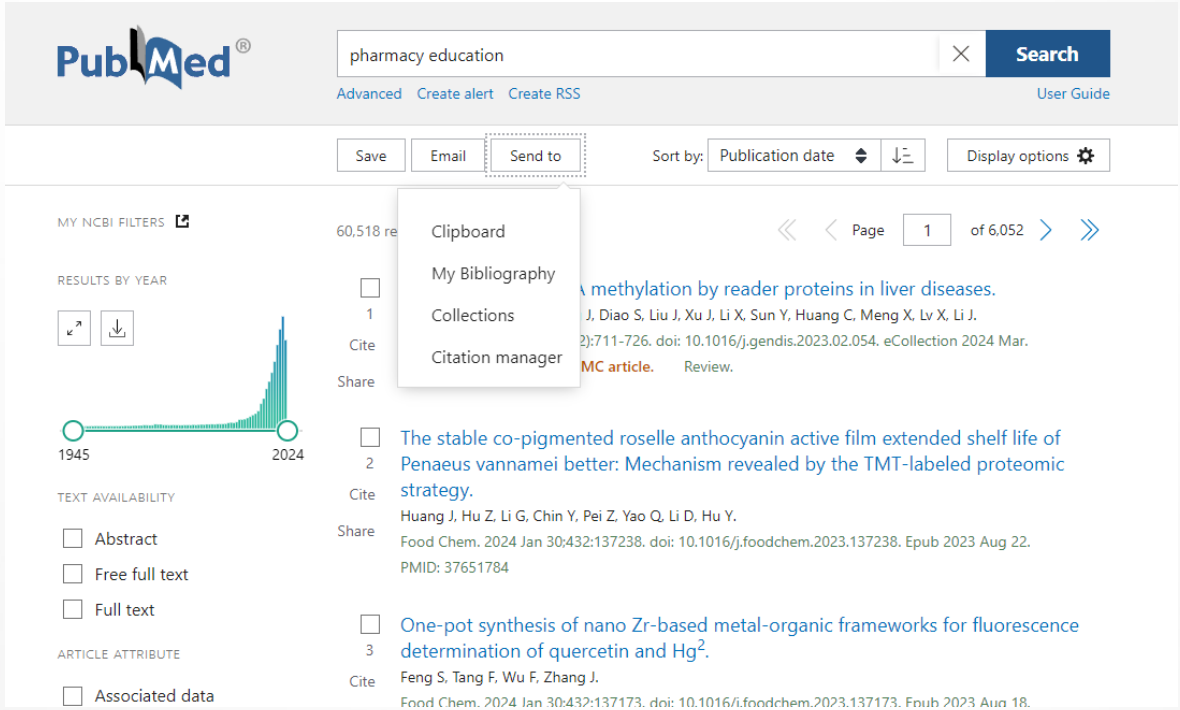


3. PubMed 결과보기

- **검색 결과 보관하기**

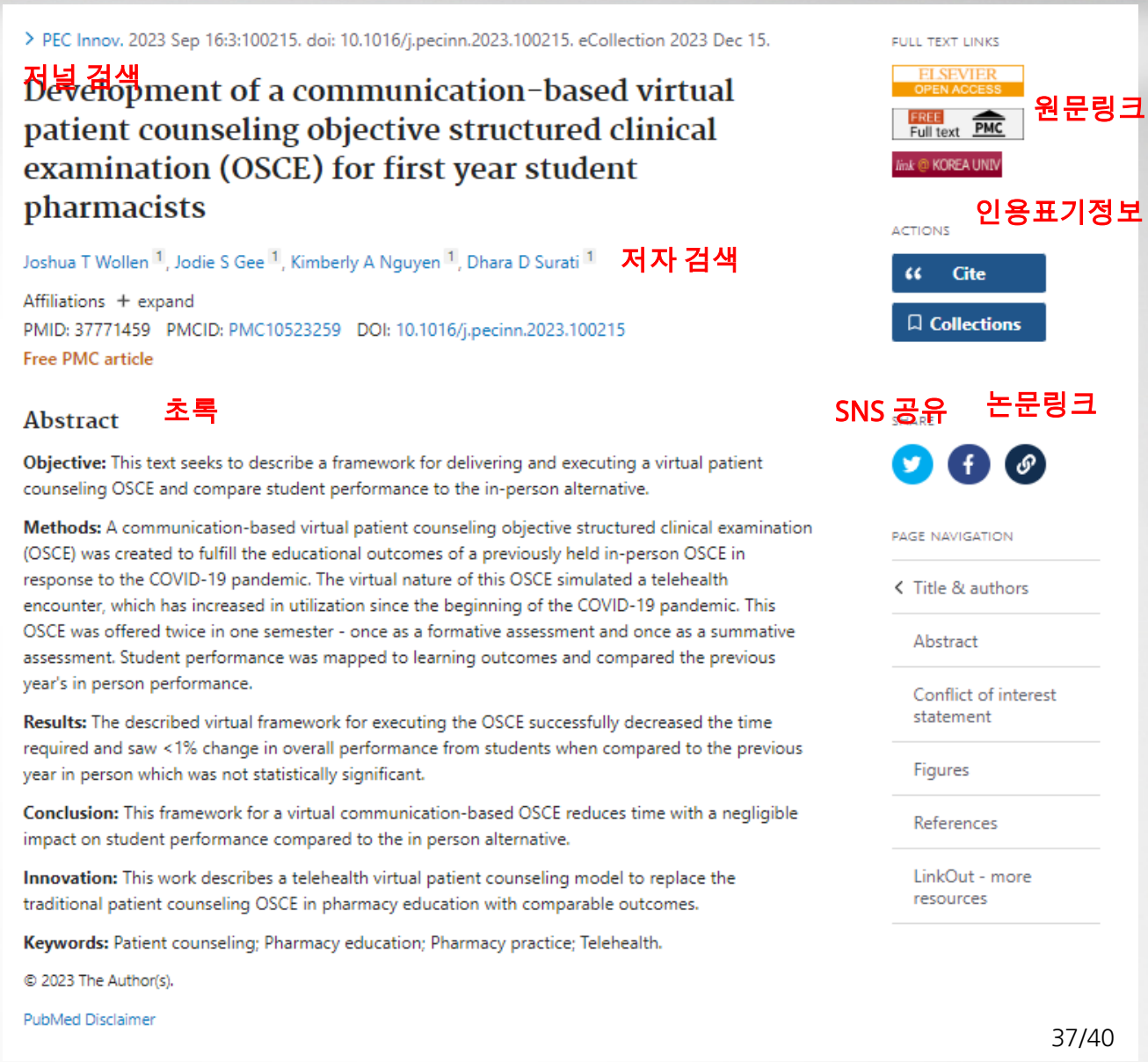
  b. Send to 기능

  - Clipboard: 임시 저장 (8시간, 최대 500건)
  - My Bibliography, Collections: 로그인 필요
  - Citation manager: EndNote 등 서지관리 프로그램



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- 상세 결과 보기



> PEC Innov. 2023 Sep 16:3:100215. doi: 10.1016/j.pecinn.2023.100215. eCollection 2023 Dec 15.

저널 검색

Development of a communication-based virtual patient counseling objective structured clinical examination (OSCE) for first year student pharmacists

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저자 검색

Affiliations + expand

PMID: 37771459 PMCID: PMC10523259 DOI: 10.1016/j.pecinn.2023.100215

Free PMC article

Abstract

초록

Objective: This text seeks to describe a framework for delivering and executing a virtual patient counseling OSCE and compare student performance to the in-person alternative.

Methods: A communication-based virtual patient counseling objective structured clinical examination (OSCE) was created to fulfill the educational outcomes of a previously held in-person OSCE in response to the COVID-19 pandemic. The virtual nature of this OSCE simulated a telehealth encounter, which has increased in utilization since the beginning of the COVID-19 pandemic. This OSCE was offered twice in one semester - once as a formative assessment and once as a summative assessment. Student performance was mapped to learning outcomes and compared the previous year's in person performance.

Results: The described virtual framework for executing the OSCE successfully decreased the time required and saw <1% change in overall performance from students when compared to the previous year in person which was not statistically significant.

Conclusion: This framework for a virtual communication-based OSCE reduces time with a negligible impact on student performance compared to the in person alternative.

Innovation: This work describes a telehealth virtual patient counseling model to replace the traditional patient counseling OSCE in pharmacy education with comparable outcomes.

Keywords: Patient counseling; Pharmacy education; Pharmacy practice; Telehealth.

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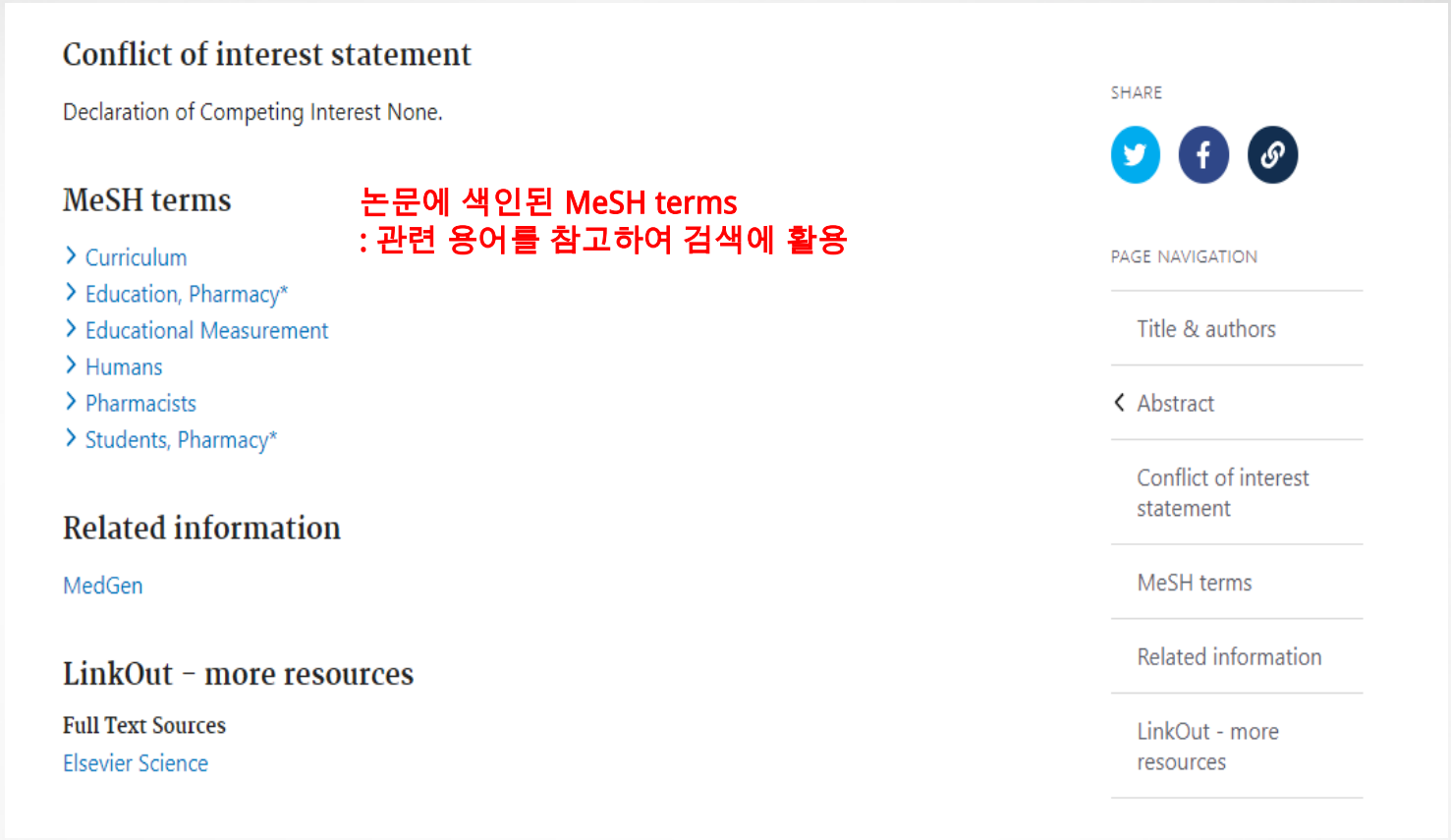
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Conflict of interest statement

Declaration of Competing Interest None.

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